



Metabolism of iodoform in dairy cows: Absorption and excretory pathways

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ABSTRACT

The objective of this study was to characterize the absorption, distribution, metabolism, and excretion (ADME) of iodoform in dairy cows through 3 complementary experiments. In study I, biological fluids (ruminal, duodenal, serum, milk, and urine) were collected from Danish Holstein dairy cows during a dose–response experiment conducted as a 4 × 4 Latin square design. The samples were used to determine the fate of iodoform and its metabolite, diiodomethane, by quantifying the amount of iodoform and diiodomethane in these samples. The cows were multi-cannulated and administered different iodoform doses (0, 320, 640, 800 mg/d) twice daily directly into the rumen. This experiment was originally designed to determine the methane-mitigating effects of iodoform, for which the methane production, milk yield, nutritional, and health status outcomes have been published separately. In the present analysis, iodoform concentrations were consistently below detection limits in all sampled matrices. However, diiodomethane was present in all matrices, though only at trace levels in milk and urine (below lower limit of quantification). To address remaining uncertainties about the ADME, additional experiments were performed. In study II, 3 rumen-cannulated Danish Holstein dairy cows received a single pulse dose of iodoform (640 mg) directly into the rumen, followed by serial blood sampling via an indwelling catheter in the jugular vein. Milk samples were collected from the subsequent milkings after dosing. Diiodomethane was above the limit of quantification in blood and milk. The apparent systemic availability of diiodomethane in blood serum ranged from 0.1% to 1.8% of the administered iodoform dose, with milk distribution at 0.06% ± 0.01% of the administered dose. In study III, an in vitro investigation of the degradation kinetics in ruminal fluid demonstrated that iodoform was rapidly metabolized (half-life [$T_{1/2}$] = 5 min), producing diio-

domethane with a slower disappearance rate ($T_{1/2}$ = 94 min). The findings from all 3 studies demonstrate that iodoform is rapidly converted to diiodomethane in rumen fluid, and the metabolite is absorbed into circulation, transferred into milk (only at high doses of iodoform), and excreted only in small amounts via urine.

Key words: methane mitigation, diiodomethane, metabolism, absorption, excretion

INTRODUCTION

Iodoform, also recognized as triiodomethane, is classified within the family of organic halogen compounds, known as halomethanes (Figure 1). Notable antimicrobial properties have led to the compound's historical use in medical settings as a disinfectant, although its application has been limited by a distinct, persistent, and unpleasant odor (Dimitriu et al., 2019). Recent research has identified iodoform as a potent inhibitor of methane when administered directly into the rumen of rumen-cannulated dairy cows, with reductions in methane yield of 15% to 75% (Thorsteinsson et al., 2023a). The chemical structure and bioactivity of iodoform have similarities to other trihalomethanes, such as bromoform, dibromochloromethane, chloroform, and bromodichloromethane (Figure 1). These compounds have been extensively studied for their toxicology and metabolic pathways in rodents and humans due to their common occurrence as byproducts in chlorinated water. (Faustino-Rocha et al., 2016).

Bromoform, a trihalomethane naturally occurring in the red marine seaweed *Asparagopsis* spp., has been shown to significantly reduce methane emissions in ruminant species (Li et al., 2018; Roque et al., 2019a, 2021; Kinley et al., 2020). Despite this mitigation potential, bromoform is classified as a potential human carcinogen (DeMarini, 2020), and concerns have been raised regarding the environmental impact of bromoform and other halogenated compounds emitted during the cultivation of seaweed, contributing to ozone layer depletion (Keng et al., 2021; Jia et al., 2022). Furthermore, visual examinations of the rumen wall and papillae of cows in certain studies have revealed concerning findings at high doses

Received July 30, 2025.

Accepted December 6, 2025.

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

of *Asparagopsis*, such as visible hemorrhages, ulcers, local spots exhibiting papillae loss, and thickening of the rumen wall (Li et al., 2018; Muizelaar et al., 2021). These outcomes emphasize the critical need for comprehensive health and safety assessments when considering the implementation of novel feed additives. A fundamental aspect of such risk evaluations involves a thorough understanding of the absorption, distribution, metabolism, and excretion (ADME) profiles of these compounds. This understanding is essential for discerning potential impacts on the well-being of the animal itself, as well as the consumers of animal-derived products.

Several studies examining halomethanes as possible methane-mitigating feed additives for dairy cows, heifers, or beef steers have tried to identify their distribution into milk or meat. Such investigations have highlighted the possible transfer of bromoform, bromine, and iodine to milk (Muizelaar et al., 2021; Stefanoni et al., 2021; Alvarez-Hess et al., 2023; Krizsan et al., 2023), and iodine into meat (Roque et al., 2021), when dairy cows and growing beef steers were fed *Asparagopsis*. In the studies by Muizelaar et al. (2021), Kinley et al. (2020), and Cowley et al. (2024), bromoform could not be detected in the meat or fat of dairy cows, steers, or heifers after treatment with *Asparagopsis taxiformis*, although Cowley et al. (2024) found increased concentrations of bromine in blood serum, kidney, and liver of heifers. One feeding trial with *Asparagopsis armata* fed to dairy cows, presented in Alvarez-Hess et al. (2024), provided an extended description of the ADME of bromoform by analyzing the presence of bromoform, dibromomethane, and bromine in milk, urine, feces, blood, and breath samples. The study demonstrated that all 3 compounds were found in blood and that bromoform and bromine were distributed to milk. Iodine was also measured in milk, but only a tendency for higher iodine concentration was found. Bromine was the only metabolite to be excreted through urine and feces, and none of the metabolites were detected in breath samples. Given the structural similarity between bromoform and iodoform, and their shared halomethane characteristics, it is essential to clarify whether iodoform follows a comparable metabolic fate in ruminants. Understanding this is particularly relevant as iodoform has been explored as an alternative halomethane with potential methane-mitigating effects without possible carcinogenic effects.

The aim of this study was to characterize the ADME profile of iodoform in dairy cattle through 3 separate experiments. The first study measured iodoform and diiodomethane concentrations in biological fluids, including ruminal, duodenal, serum, milk, and urine samples, collected from a dose-response feeding trial. Although this experiment was originally designed to assess the methane-mitigating effects of iodoform in dairy cows (Thorsteinsson et al., 2023a), the present study provides

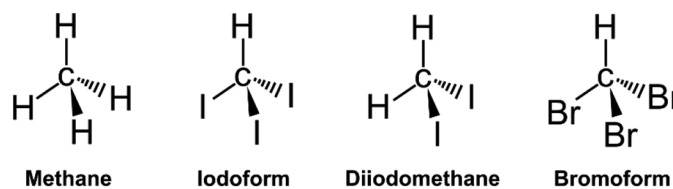


Figure 1. Molecular structure of methane and the halogenated compounds iodoform, diiodomethane, and bromoform.

novel insights into the compound's ADME profile, which was not addressed in the previous report. Due to some inefficiency of the ADME data from the dose-response trial, 2 follow-up independent studies were designed specifically for the ADME determination. The second study investigated the absorption kinetics of iodoform using 3 rumen-cannulated dairy cows that received a single pulse dose directly into the rumen, with serial blood samples collected via indwelling jugular vein catheters. The third study was conducted in vitro to assess the degradation kinetics of iodoform in buffered ruminal fluid.

MATERIALS AND METHODS

Biological Samples from Animal Experiments

The animal experiments were conducted at Aarhus University Viborg Research Centre Foulum (Tjele, Denmark) and complied with the guidelines set out by the Danish Ministry of Environment and Food with respect to animal experimentation and care of animals under study (Act No. 2028; Danish Ministry of Environment and Food, 2020).

Study 1: In Vivo Dose-Response Experiment. For the present ADME study, biological samples, including ruminal fluid, duodenal fluid, blood serum, urine, and milk, were obtained from a previously conducted dose-response 4×4 Latin square experiment (Thorsteinsson et al., 2023a) involving 4 Danish Holstein dairy cows, each fitted with rumen, duodenal, and ileal cannulas. This experiment was initially designed to examine the methane-mitigating effects of iodoform, and details regarding the animals, housing, diet, and experimental procedures have been reported in the cited study. In short, the experiment consisted of 4 consecutive 2-wk periods. In the 4 experimental periods, multiparous cows (2 second-parity and 2 third-parity; BW: 666 ± 60 kg; milk yield: 33.1 ± 2.10 kg/d; DIM: 210 ± 66 d; DMI: 23.0 ± 1.69 kg/d) were fed the same TMR consisting of maize silage and grass-clover silage (spring growth and first regrowth) as roughage, and soybean meal, sugar beet pulp, rapeseed cake, barley, and mineral supplements as concentrates. The cows were milked and fed twice daily at 0630 h and 1630 h. The doses of iodoform were 0, 320, 640, and 800 mg/d, and half of the daily dose was provided di-

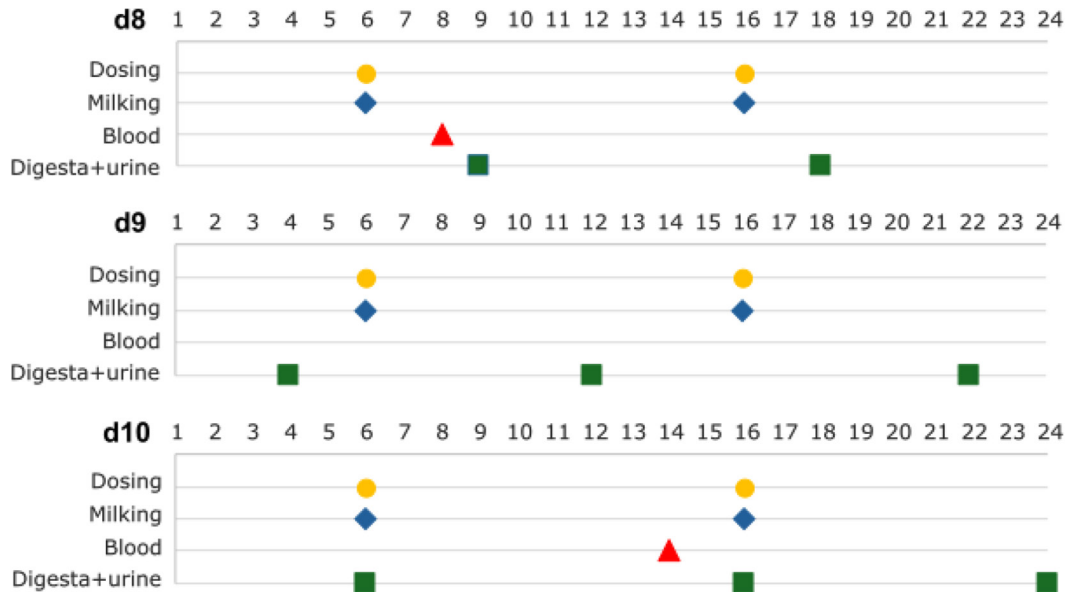


Figure 2. Timeline depicting sampling times at d 8, 9, and 10 in each of the 4 experimental periods in study I, in which dairy cows received intraruminal administration of pulse doses of iodoform. Iodoform was administered in the rumen at each feeding (●; yellow). Milk samples (◆; blue) were taken at each of the daily milkings (0630 h and 1630 h), corresponding to 10 h and 14 h after the last iodoform dosing. Blood samples (▲; red) were taken 2 h and 8 h after iodoform dosing on d 8 and d 10. Ruminal fluid, duodenal fluid, and urine (■; green) were all sampled at 9-h intervals, corresponding to time points from 2 h to 14 h after iodoform dosing.

rectly into the rumen at each feeding. These doses were based on a previous unpublished pilot study. The half-daily dose of iodoform was dissolved in 20 mL of 99% ethanol and mixed with 0.5 kg of pelleted concentrate mixture before being administered into the rumen. During d 8 to d 10 of each experimental period, rumen fluid, duodenal content, and urine samples were collected at 9-h intervals to represent every third hour of a 24-h period and represent a time frame from 2 to 14 h after iodoform administration (Figure 2). Milk samples were collected twice daily at each milking on sampling days, corresponding to 10 h and 14 h after the morning and afternoon iodoform treatment, respectively (Figure 2). Two blood samples were collected in serum evacuated tubes (Greiner Bio-One GmbH, Kremsmünster, Austria) during each experimental period at d 8 and d 10 by venipuncture of the tail artery or vein, corresponding to 2 h and 8 h after the morning administration of iodoform, respectively (Figure 2). The blood samples were left to coagulate for at least 1 h at room temperature before being centrifuged at $1,300 \times g$ at 20°C for 10 min. Afterward, the serum was transferred to cryotubes. All samples were stored at -80°C for ~ 1 yr before GC-MS/MS analysis. Given this extended storage period, the possibility of volatilization or degradation cannot be excluded. Moreover, it cannot be ruled out that some iodoform and diiodomethane may have been broken down during freezing and thawing.

As described in Thorsteinsson et al. (2023a), the highest iodoform dose was initially intended to be 1,080 mg/d, but

due to a substantial reduction in DMI, the dose needed to be adjusted after the first experimental period, and only data for the 3 cows that received the treatment 800 mg/d in the subsequent 3 experimental periods are included in the quantification of iodoform and diiodomethane.

Study II: In Vivo Absorption Kinetic Study with Dairy Cows. Three multiparous Danish Holstein dairy cows (one second parity and 2 third parity; BW: 655 ± 27 kg, milk yield: 24 ± 2 kg milk/d; DIM: 210 ± 146 d) were fitted with a temporary indwelling jugular catheter (1.02 mm i.d., 1.78 mm o.d.; Tygon S-54-HL, Buch & Holm A/S, Herlev, Denmark) ~ 1 h before first blood sampling. The catheter was inserted 20 cm into the right external jugular vein by percutaneous venipuncture using a hypodermic needle (2.5×110 mm, Medioplast AB, Malmö, Sweden) after shaving, skin disinfection, and subcutaneous injection of 2 mL of xylocaine (Lidocaine 20 mg/mL, AstraZeneca A/S, Albertslund, Denmark). The catheter was secured to the skin with suture after removal of the hypodermic needle and kept in a pocket made from animal polster Snøgg (Norway Plaster AS, Vennesla, Norway; 15×15 cm; E-vet A/S, Haderslev, Denmark) and additionally secured with Kruuse Vet-Flex (Kruuse A/S, Langerskov, Denmark). The catheter was flushed after each blood sampling with 10 mL of heparinized saline (10 IU/mL). Blood samples were collected at time points 0, 15, 30, 45, 60, 120, 240, and 360 min after 640 mg of iodoform (dissolved in 25 mL of ethanol [99%] and mixed with 0.5 kg of concentrate

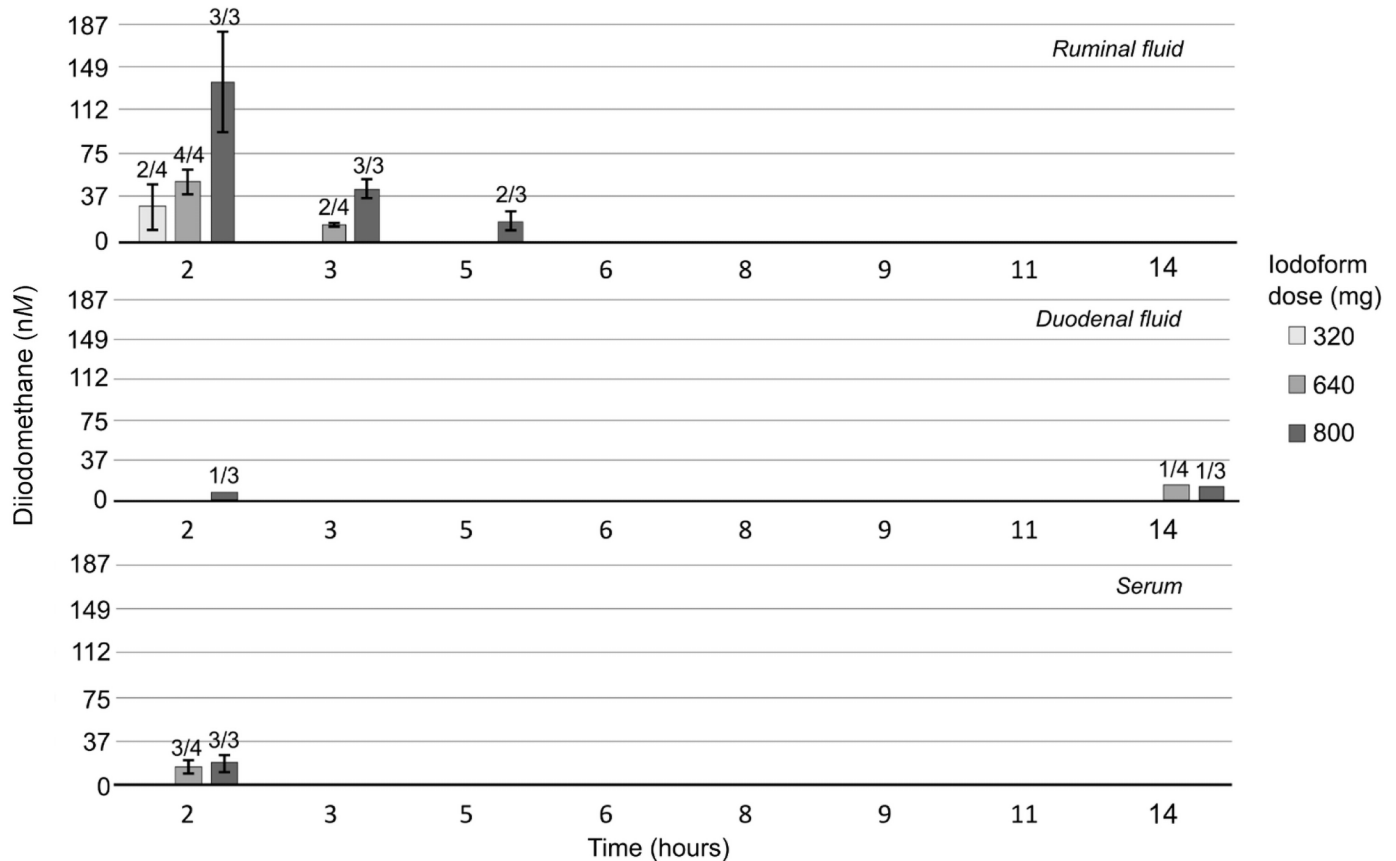


Figure 3. Concentration (mean $\pm$ SD of means) of diiodomethane; the number of samples with detectable diiodomethane (<7.5 nM) is shown above each bar. The different matrices are ruminal fluid, duodenal fluid, and blood serum collected from 4 cows with administration of iodoform directly into the rumen through a rumen cannula in different doses (0, 320, 640, 800 mg/d) in a 4×4 Latin square design (study I). The number of samples is 4 at each time point, except for dose 800, where data are missing from 1 cow. Time points 3, 6, and 9 h are after the morning dosing, whereas 2, 5, 8, 11, and 14 h are after the evening feeding. Diiodomethane was not present in any samples from experimental periods where the cows received 0 mg/d of iodoform and is not shown in the figure.

pellets) was administered directly into the rumen. After removing the heparinized saline in the infusion line (clear liquid), the first 2 mL of blood collected in a syringe (10 mL) were discarded. Another syringe (10 mL) was used to sample blood, which was afterward transferred to serum evacuated tubes (Greiner Bio-One GmbH, Kremsmünster, Austria), allowed to coagulate at room temperature for at least 1 h, and then centrifuged at $1,300 \times g$ at 20°C for 10 min. Milk samples were additionally collected at 4 milkings after the iodoform intraruminal dosing (5, 17, 26, and 41 h after dosing) to determine the amount of excretion of iodoform and diiodomethane to milk. All samples were stored at -80°C before iodoform and diiodomethane quantitation on GC-MS/MS within few days after collections.

Study III: In Vitro Rumen Degradability of Iodoform and Diiodomethane. For this in vitro study, the procedure for rumen fermentation studies described by Thorsteinsson et al. (2023b) was used. In short, ruminal fluid was col-

lected 30 min before the morning feeding from 3 rumen-cannulated nonlactating (dried off after first lactation) and nonpregnant Danish Holstein cows (BW: 805 ± 28 kg). The cows were fed a diet fulfilling maintenance requirements consisting of grass-clover hay, barley straw, and a pelleted concentrate mixture (40% barley grain, 40% oat grain, 10% soybean meal, 3% rapeseed meal, 3% sugar beet molasses, and 4% commercial mineral mixture per kilogram of fresh mixture). The diet was formulated according to the Nordic Feed Evaluation System (Norfor; Volden, 2011) using the “NorFor Feed Ration Optimizer” software.

The ruminal fluid was filtered through one layer of moist cheesecloth at the laboratory. Buffer, redox indicator, reducing agents, and macro and micro mineral solution were prepared as described in Menke and Steingass (1988) and kept under anaerobic conditions by continuously flushing with N_2 until the addition of ruminal fluid in a 2:1 ratio of buffer solution to ruminal fluid. Next, 90 mL of inoculum was dispensed into Duran bottles (capacity: 132 ± 1.1 mL;

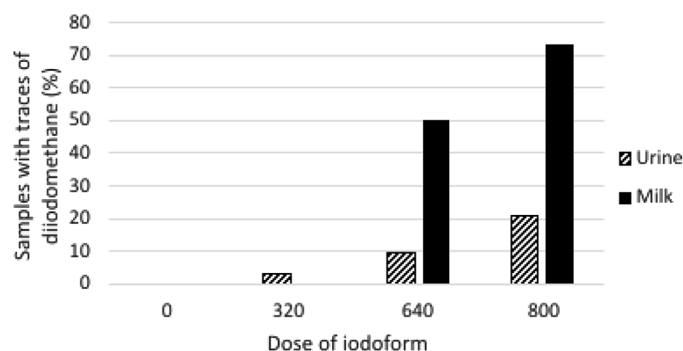


Figure 4. Percentages of milk samples where trace amounts (<7.5 nM) of diiodomethane were detectable, when cows had been given different intraruminal doses of iodoform twice daily (for doses 0, 320, 640: n = 16; for dose 800 mg/d: n = 15) and urine (for each of the doses 0, 320, 640 mg/d: n = 32; for 800: n = 24) from 4 dairy cows that received 4 different doses of iodoform (0, 320, 640, 800 mg/d) directly in the rumen in a 4 × 4 Latin square design (study I). The highest dose was adjusted in the first experimental period, meaning that dose 800 mg/d is missing from 1 cow.

DWK Life Sciences, Mainz, Germany) before adding 100 µL of iodoform or diiodomethane in an ethanol solution (35 µg/mL). The headspace of the bottle was flushed with N₂ before an Ankom pressure sensor module (Ankom Technology, Macedon, NY) was attached. Bottles were incubated at 38.5°C for 0, 1, 2, 5, 10, 15, 20, 25, 30, 45, 60, 120, 180, 240, 360, or 1,440 min in a controlled incubator shaker (New Brunswick Excella E25R, Eppendorf, Hamburg, Germany) with oscillation of 50 rpm. Gases in the headspace of the bottles were automatically released from the bottles when a pressure of 5.17 kPa was reached. Three bottles without any added iodoform or diiodomethane were included as controls and incubated for 0, 360, and 1,440 min. Fermentation was stopped by placing the bottles on ice, after which 2 replicate samples of 1 mL were collected from each bottle and immediately treated with 1 mL of methanol to achieve an instant stop in fermentation. The samples were stored at -80°C before being analyzed for contents of iodoform and diiodomethane on GC-MS/MS within a few days after collection.

Chemicals for GC-MS Analyses

For GC-MS analyses, the following standards were used: iodoform (Qmx Laboratories, Essex, UK), diiodomethane (Merck-Sigma A/S, Søborg, Denmark), and diiodomethane-d₂ (Merck-Sigma A/S; internal standard). Methanol and dichloromethane were purchased as SupraSolv for GC-MS (Merck-Sigma A/S). Diiodomethane and diiodomethane-d₂ (99% purity) and the iodoform standard were dissolved to 1,000 µg/mL in methanol. Working solutions were prepared in concentrations of 1,000 ng/mL in methanol and used for preparation of

standard curves and quality control samples. The working solutions and the standard solutions were stored at -20°C when not in use.

Instrumentation

The GC-MS system used for the analysis was a Trace 1310 gas chromatograph with a TSQ 9000 quadrupole mass spectrometer (Thermo Fisher A/S, Roskilde, Denmark). We used an Rxi-5ms column (low polarity phase, Crossbond diphenyl dimethyl polysiloxane; 30-m length, 0.25-mm i.d., 0.25-µm film thickness; WVR International A/S, Søborg, Denmark). The flow rate of the carrier gas (He) was 1.2 mL/min with sample injection using a split ratio of 1.5:10. The injection volume was 1 µL. The inlet and transfer line temperatures were 250°C and 280°C, respectively. The temperature program for the column was as follows: initial temperature was 40°C (5-min hold), followed by an increase to 160°C (1-min hold) at 20°C/min and a second increase to 280°C (1-min hold) at 60°C/min. This program used 18 min plus 1 to 2 min of cool down per injection. The ionization of iodoform and diiodomethane was achieved by electron impact (70 eV) using single reaction monitoring mode.

Liquid-Liquid Extraction

Iodoform and diiodomethane were extracted from biological samples using a previously developed method for ruminal, duodenal, milk, and urine matrices from cattle (Rønn et al., 2025). Method development demonstrated that the quantification approach achieved high recovery rates for both compounds across all matrices when liquid-liquid extraction (LLE) was employed for phase separation into a dichloromethane phase. For quantification, LLE consisted of a 1:1:1 mixture of dichloromethane, methanol, and sample. Urine, ruminal, and duodenal fluids were centrifuged at 14,000 × g at 20°C for 10 min before extraction. For the extraction, 1 mL of sample and 1 mL of methanol with internal standard (50 ng/mL) were added to a 15-mL Greiner tube. The methanol-sample solution was lightly mixed before 1 mL of dichloromethane was added, then vortexed for 10 min followed by 10 min of centrifugation at 3,500 × g and 20°C to obtain a phase separation.

Standard Curves

Standard solutions were prepared in the range of 1 to 100 ng/mL in methanol with an internal standard concentration of 50 ng/mL, with standard curves constructed using Chromeleon CDS Software (7.3, 2023, Thermo Fisher A/S). The analyte/internal standard concentration ratio was plotted against the analyte/internal standard

Table 1. Rate (k) and half-life ($T_{1/2}$) of iodoform (IF) and diiodomethane (DIIM) calculated from linear models with logarithmic transformation of concentration as response variable and time from $C_{\max}$ as fixed effect

Study ¹	Site	Compound	k	$T_{1/2}$ (min)	R^2
Study I	Rumen	DIIM	0.6770	61.4	0.80
Study II	Blood serum	DIIM	0.0044	157.5	0.69
Study IIIa	Ruminal fluid	IF	0.1333	5.2	0.90
		DIIM	0.0074	93.7	0.98
Study IIIb	Ruminal fluid	DIIM	0.0021	330.1	0.97

¹Study I: In vivo disappearance of diiodomethane from the rumen in 4 dairy cows that had been given an intraruminal bolus dose of iodoform. Study II: In vivo metabolism and disappearance of absorbed diiodomethane in 3 dairy cows that had been given an intraruminal bolus dose of iodoform. Study IIIa: In vitro disappearance of iodoform and diiodomethane formed after incubation of iodoform in buffered ruminal fluid. Study IIIb: In vitro disappearance of diiodomethane after incubation in buffered rumen fluid.

peak area ratio as a linear regression curve with $1/x$ weighting. Based on a validated method for iodoform and diiodomethane analyses (Rønn et al., 2025), the lower limit of quantification values for iodoform and diiodomethane are 5.1 nM and 7.5 nM, respectively, whereas the lower limit of detection (LLOD) was estimated by visual validation of chromatograms.

Statistical Analysis

All statistical analyses were handled in R 4.3.1 (R Core Team, 2023). For study I, the rate of degradation, absorption, and metabolism of diiodomethane in ruminal fluid was determined with linear models using the highest measured concentrations ($C_{\max}$) with logarithmic transformation of concentration as the response variable and time as the predictor variable using the lme package in R (Bates et al., 2015) with cow as a random effect. Only cows receiving the highest dose of iodoform (800 mg) were used for these calculations to have enough time points for a linear model (3 time points available from 3 cows). Half-life ($T_{1/2}$) was calculated based on the following formula:

$$T_{1/2} = \frac{\ln(2)}{k}, \quad [1]$$

where k is the elimination rate of the log-transformed linear model.

For study II, to estimate the total area under the curve (AUC) from time zero to infinity ($AUC_{0-\infty}$), the AUC from the last measurable concentration (AUC_{0-t}) was extended to infinity ($AUC_{0-\infty}$) by extrapolating the terminal elimination phase (CDER, 2021). This extrapolation assumes that the metabolite follows a monoexponential decay during the terminal phase, characterized by a constant elimination rate constant (k). The $AUC_{0-\infty}$ is calculated using the following formula:

$$AUC_{0-\infty} = AUC_{0-t} + \frac{C_{\text{last}}}{k}, \quad [2]$$

where C_{last} is the last measurable serum concentration and k is the elimination rate constant. The rate of metabolism or elimination (k) was estimated using linear regression on the natural logarithm of the serum concentrations during the terminal phase (from $C_{\max}$) with cow as random effect ($n = 3$). The $T_{1/2}$ was calculated as described in Formula [1].

To calculate the bioavailability, an assumption of volume of distribution (V_d) was made based on the characteristics of diiodomethane. Given its low water solubility and organic solvent solubility (Sigma-Aldrich, 2024), diiodomethane could be distributed into lipid-rich tissues. Values for diiodomethane in blood could not be found; however, dichloromethane, another dihalomethane, has been reported to exhibit a V_d of 0.6 to 1.5 L/kg in rodents (EPA, 1985). In pharmacologic approaches, the default assumptions state that compounds that distribute into tissues and fat have a high V_d of ≥ 1 L/kg (Robinson et al., 2025). Based on these assumptions, 3 V_d values were computed at 0.6, 1.0, and 2.0 L/kg to estimate the sensitivity and a likely range of systemic availability of diiodomethane from administered iodoform. The distribution of diiodomethane to milk was calculated based on each individual cow's milk production and the measured diiodomethane concentration at the first milking after iodoform administration.

In study III, for the in vitro degradation analysis of iodoform and diiodomethane, the $T_{1/2}$ was estimated using a linear regression model. For diiodomethane in ruminal fluid, the final measurement at 24 h was excluded because the compounds were incubated without substrate, and under these conditions, a decline in microbial activity could not be ruled out.

RESULTS

Study I: In Vivo Dose-Response Experiment with Dairy Cows

Iodoform concentrations in all biological matrices (ruminal fluid, duodenal fluid, blood serum, urine, and milk) were below the LLOD. Diiodomethane was measurable in ruminal fluid, duodenal fluid, and blood serum (>7.5

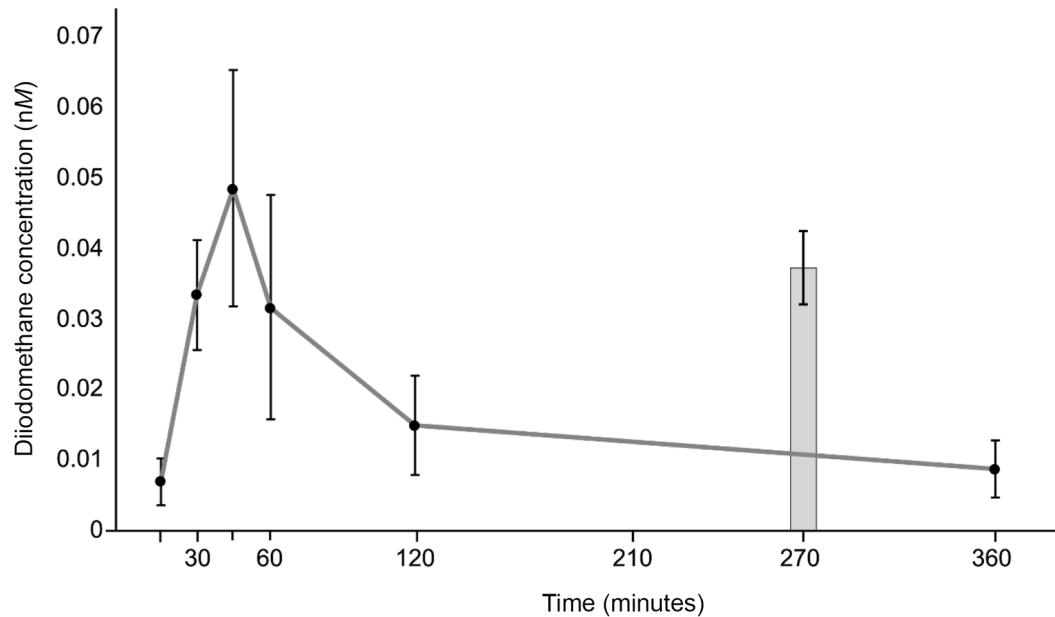


Figure 5. Average concentration (mean $\pm$ SD of the mean) of diiodomethane in blood serum (black dots and lines) and milk (gray bar) after a pulse dose of iodoform (640 mg) was administered directly into the rumen of 3 rumen-cannulated dairy cows (study II).

nM; Figure 3) and detected at trace levels in milk and urine (<7.5 nM; Figure 4). The low or absent concentrations of iodoform and diiodomethane indicated that the limited sampling scheme in the dose–response study was insufficient to fully characterize the ADME profile of iodoform, hence the necessity to expand this study with study II and study III.

The concentration of diiodomethane was affected by iodoform dose (treatment) and by the time after the last iodoform dosing in all matrices. In ruminal fluid, it was possible to detect diiodomethane across all the 3 iodoform treatments, but only in the first collected sample (2 h after dosing) for the 320 mg/d dose, the first 2 samples (2–3 h after dosing) for the 640 mg/d dose, and the first 3 samples (2–5 h after dosing) for the 800 mg/d dose. For duodenal fluid, only a few samples from the 640 and 800 mg/d doses had measurable diiodomethane levels at 2 h and 14 h after iodoform dosing. For the blood serum, the samples were only collected at 2 different time points, and diiodomethane was present only in samples from cows on the 2 highest treatments (640 and 800 mg/d) and only in the sample taken 2 h after iodoform dosing, and not in the sample collected 8 h after dosing. For urine and milk, the concentrations of diiodomethane were below the quantification limit ($<$ LLOD), although diiodomethane was detectable in some samples. The percentage of urine samples with traces of diiodomethane increased with increasing iodoform supplementation (Figure 4), and in milk, we found a considerable increase in proportion of samples with detectable levels of diiodomethane

from 0% to 50% to 73% for the 320, 640, and 800 mg/d doses, respectively. Diiodomethane had a $T_{1/2}$ of 61.4 min ($R^2 = 0.80$) in the rumen (Table 1).

Study II: In Vivo Absorption Kinetic Study with Dairy Cows

Similar to study I, it was not possible to detect iodoform in any blood samples, despite the commencement of sampling as early as 15 min after iodoform dosing. In contrast, for diiodomethane, peak concentrations were observed between 30 and 45 min after iodoform dosing (Figure 5). By 6 h after dosing, diiodomethane was quantifiable (>7.5 nM) in the blood of 1 out of 3 cows, and by the final blood collection at 8 h after dosing, it was below the LLOD in all cows. The $T_{1/2}$ of diiodomethane in blood serum was estimated to be 157.5 min ($R^2 = 0.69$; Table 1). Based on this, the systemic availability of diiodomethane in blood serum from administered iodoform was calculated to have an apparent range between 1.5% and 4.8% (Table 2).

Iodoform was not detectable in any of the analyzed milk samples ($<$ LLOD). However, in the first milk sample collected 270 min after iodoform dosing, the average diiodomethane concentration was recorded at 46.4 ± 6.6 nM (Figure 5). Subsequent milk samples collected from 15 h to 41 h after iodoform dosing did not have detectable levels of diiodomethane. The average amount of diiodomethane found in milk was 0.24 ± 0.05 mg based on the milk yield of the cows, corresponding to an average distribution of

Table 2. Apparent systemic availability of diiodomethane (DIIM) derived from iodoform (IF) in blood serum, estimated using an in vivo kinetic study in which 640 mg of IF was administered directly into the rumen of 3 rumen-cannulated dairy cows¹

Vd ² (L/kg)	C _{max} (nM)	T _{1/2} ³ (min)	CL (mL/min)	AUC _{0-∞} (nM/min)	DIIM systemic mass (μmol)	Percentage of IF dose ⁴
0.6	58 ± 18	157.5	1,730 ± 58	5,417 ± 2,984	23.7 ± 9.5	1.5 ± 0.6
1.0			2,883 ± 97		38.9 ± 15.9	2.4 ± 1.0
2.0			5,765 ± 194		77.8 ± 31.9	4.8 ± 2.0

¹The estimation of apparent systemic availability (IF → DIIM) was conducted under 3 alternative assumptions regarding the volume of distribution (Vd) of DIIM in blood, thereby providing a plausible range of systemic availability values for dairy cows. Data are reported as mean ± SD, except T_{1/2}. CL = clearance, C_{max} = peak concentration, T_{1/2} = half-life, AUC_{0-∞} = area under the curve from time 0 to infinity.

²Assumptions based on values in EPA (1985) and Robinson et al. (2025).

³Based on linear regression with cow (n = 3) as random effect. R² = 0.69.

⁴Calculated by moles of IF found in blood serum as moles of DIIM in percentage.

0.06% ± 0.01% of the administered iodoform dose to milk. This calculation assumes that all circulating diiodomethane in the blood was absorbed, distributed, metabolized, and excreted before the first milking.

Study III: In Vitro Rumen Degradability of Iodoform

The in vitro rumen fermentation study demonstrated that iodoform disappears almost instantly when added to ruminal fluid (Figure 6a). The T_{1/2} of iodoform was 5.2 min (R² = 0.90; Table 1); thus, no iodoform could be detected after 30 min of incubation in ruminal fluid. Upon disappearance of iodoform, diiodomethane was formed reaching a C_{max} of 45.3 nM after 15 min of rumen fermentation, after which diiodomethane concentrations decreased with a T_{1/2} of 93.7 min (R² = 0.98). In contrast to the low T_{1/2} of iodoform, diiodomethane had a T_{1/2} of 330.1 min (R² = 0.97) when the same amount of diiodomethane was added to the ruminal fluid and was still detectable after 24 h of in vitro rumen fermentation (Figure 6b).

DISCUSSION

Iodoform Metabolism

Based upon the conditions of these experiments, the metabolism of iodoform appears to be initiated in the rumen, possibly involving rapid dehalogenation microbes. This rapid degradation, with a T_{1/2} of 5.2 min in ruminal fluid (Table 1), explains why iodoform was undetectable in any sample from the 4 × 4 feeding trial (study I) and in the blood serum even as early as 15 min after ruminal dosing in the in vivo kinetic study (study II). This suggests that only a negligible amount, if any, is absorbed from the rumen or intestines into the blood. Halomethanes are also metabolized in the liver (Gargas et al., 1986), which may further explain the absence of detectable iodoform in blood. During the dehalogenation in the rumen, diiodomethane is formed and subsequently absorbed into the bloodstream. Iodomethane could not

be measured using the GC-MS/MS quantification method, although it is likely that it could be formed as well. Given its low boiling point at 42.5°C (PubChem, 2004), iodomethane might evaporate and escape from the rumen through eructation.

Methanogens have demonstrated the capacity to effectively metabolize trihalomethanes in a reductive dehalogenation to methane and other less halogenated intermediates (Krone et al., 1989a,b). Romero et al. (2023) reported that bromoform degradation in ruminal fluid is unaffected by diet, suggesting that fermentation rate is not the key factor for halomethane degradation. Instead, they proposed that degradation could be limited by depletion of cofactor F₄₃₀, which is an essential component in the final step of methanogenesis (Thauer, 2019) and required for reductive dehalogenation by methanogens or heterotrophic anaerobes. In contrast, Posman et al. (2025) argued that F₄₃₀ depletion is unlikely given the low rate of dibromomethane loss and instead suggested that low transformation yields from bromoform are more probable. In addition, nonenzymatic degradation by anaerobic bacteria, such as acetogenic and sulfate-reducing bacteria, has also been proposed to contribute to dehalogenation (Egli et al., 1990; Fetzner, 1998).

The rate of dehalogenation decreases for each halogenated intermediate removed but increases with the dissociation energies of the halogen-carbon bonds, which in turn decreases in the order of Cl > Br > I (Glockler, 1959). Thus, it can be speculated trihalomethanes with iodine will be dehalogenated more efficiently than the other 2 halogens, and trihalomethanes have a higher rate of dehalogenation than dihalomethanes. This possibly explains the rapid reduction in iodoform in ruminal fluid (Figure 6) with the formation of the metabolite, diiodomethane. Other studies looked at the degradation of bromoform in ruminal fluid in vitro and found that bromoform likewise has a fast degradation to dibromomethane (Romero et al., 2023; Posman et al., 2025). However, the degradation of bromoform and dibromomethane was slower at 26 min and 775 min, respectively, than in the current study, following the dissociation energies

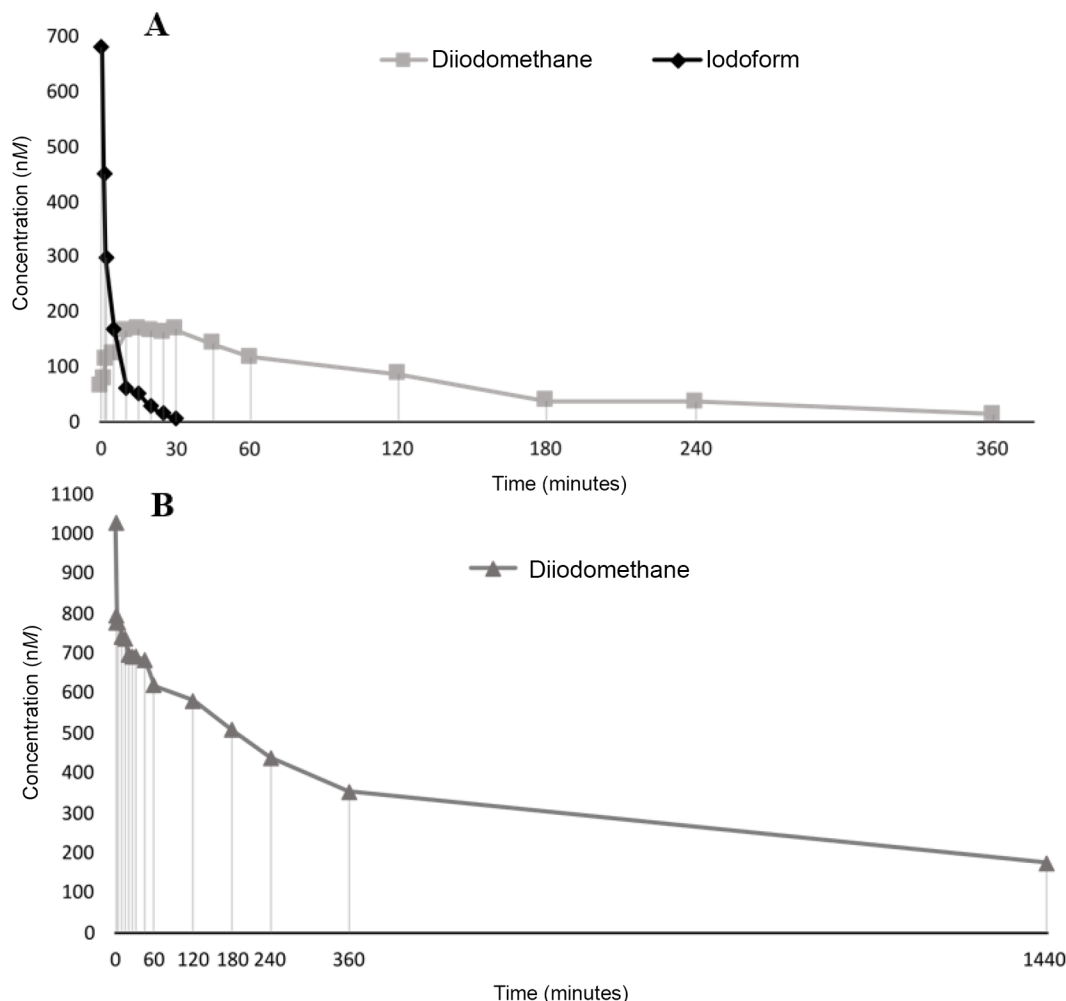


Figure 6. Study III. (A) In vitro degradation of iodoform (◆) and the associated formation and subsequent disappearance of diiodomethane (■) in buffered ruminal fluid following the addition of 35 µg/mL iodoform at time zero. Concentrations reflect fermentation stopped at each sampling time point. (B) In vitro degradation of diiodomethane (▲) in buffered ruminal fluid following the addition of 35 µg/mL diiodomethane at time zero, with fermentation stopped at each sampling time point.

of iodine versus bromine. The difference between the observed $T_{1/2}$ of diiodomethane in ruminal fluid in vivo in study I and the calculated $T_{1/2}$ determined in vitro of diiodomethane in the rumen in study III (Table 1) might also be affected by absorption from the rumen and passage rate of diiodomethane with rumen contents out of the rumen, which reduces the actual $T_{1/2}$ of the compound in the rumen in vivo by ~30 min. Based on the concentrations measured in blood early during iodoform dosing and low concentrations measured in duodenum, the absorption rate of diiodomethane is likely greater than the passage rate out of the rumen.

Excretion Pathways

Other studies on chloroform and bromoform, along with the equilibrium observed between human blood,

urine, and milk, suggest a preference for milk rather than blood, and blood rather than urine, due to the lipophilic nature of trihalomethanes (Batterman et al., 2002). The rapid increase in the percentage of milk samples with traces of diiodomethane with the increase in daily dose of iodoform (0%, 50%, and 73% with the 320, 640, and 800 mg/d doses, respectively; Figure 4) shows that diiodomethane is excreted through milk. This is further evident in the low number of urine samples with traces of diiodomethane (Figure 4), where diiodomethane was detected in only a few urine samples in study I, indicating either sufficient endogenous detoxification pathways before excretion or that urine is not the main route of excretion for diiodomethane. In the study by Alvarez-Hess et al. (2024), dibromomethane was not detected in urine, feces, milk, or breath samples, suggesting either complete metabolism with liberation of free bromine or

distribution of the dihalomethane in adipose or muscle tissue. However, the deposition of the dihalomethane bromochloromethane in liver, kidney, fat, and meat was not observed in steers (Tomkins et al., 2009). Despite the fact that tri- and dihalomethanes are lipophilic and could potentially be transported and stored in adipose or muscle tissue, no study has demonstrated this occurrence (Muizelaar et al., 2021; Cowley et al., 2024). This absence of accumulation may indicate effective detoxification processes or the volatility of these compounds, preventing their accumulation in tissue. Based on these observations, there are no indications that accumulations of iodoform and diiodomethane would be expected in the tissues of dairy cows administered with these compounds in their diet.

The calculation of the apparent total availability of diiodomethane in study II (1.5%–4.8% of the administered iodoform; Table 2) demonstrates that the ruminal dose of iodoform produced a measurable systemic distribution of diiodomethane. However, these estimates are approximate and based on a limited dataset (3 cows, $R^2 = 0.69$) and several stated assumptions. Despite this uncertainty, they offer a useful order-of-magnitude indication of systemic absorption, supporting the interpretation that diiodomethane enters the circulation and may contribute to a possible transfer to other tissues. Despite its lipophilic properties, the low proportion of diiodomethane recovered in milk ($0.06\% \pm 0.01\%$ of administered iodoform) shows that some distribution to milk does occur, although at a minimal level. In the study by Alvarez-Hess et al. (2024), it was reported that feeding *Asparagopsis* to dairy cows resulted in quantifiable levels of both bromoform and dibromomethane in blood, but only bromoform in milk. This finding is unexpected given that the measured concentrations of dibromomethane in blood was higher ($5.8\text{--}47.0\text{ }\mu\text{g/L}$) compared with bromoform ($2.6\text{--}4.2\text{ }\mu\text{g/L}$). With the lipophilic nature of both these compounds, it would facilitate a transfer of both dibromomethane and bromoform to milk. These findings would suggest that trihalomethanes, rather than dihalomethanes, are more readily distributed to milk, and could explain the low transfer of diiodomethane to milk even when the cows received a high dose of iodoform.

The method of administering halomethanes influences the level of transfer to milk in cows. When iodoform was supplemented directly into the rumen 2 times per day in study I, diiodomethane was detectable at trace levels in milk ($<7.5\text{ nM}$). Similarly, in study II, one dose of 640 mg/d of iodoform into the rumen resulted in quantifiable levels of diiodomethane in the first milking after dosing (46.4 nM diiodomethane; Figure 5) and below the LLOD in the next. However, when iodoform was mixed into the TMR in Thorsteinsson et al. (2025), the milk did not have detectable levels of diiodomethane. Likewise, detect-

able levels of bromoform in milk were only seen under experimental conditions, where the feeding of *Asparagopsis* occurred at separate time points from the feeding of the TMR (Muizelaar et al., 2021; Alvarez-Hess et al., 2023, 2024), whereas studies with no significant findings of bromoform in milk involved mixing *Asparagopsis* into the TMR (Roque et al., 2019b; Stefenoni et al., 2021; Krizsan et al., 2023). These findings suggest that administering halomethanes in one large dose increases the percentage-wise escape from dehalogenation and detoxification pathways, facilitating an increases likelihood of absorption and subsequent transfer to milk. Furthermore, the inclusion of *Asparagopsis* as an oil extract resulted in substantial increases in bromoform in milk, likely because the oil format prevents dehalogenation by microbes in the digestive tract or facilitates a more rapid transfer to milk through lipid metabolism and distribution (or both; Alvarez-Hess et al., 2023, 2024). These differences in transfer rates imply that feeding management and the method of halomethane administration (whether as pure compound, from plant material, or as part of an oil extract) can influence the ADME behavior of these compounds.

Iodine Distribution

Although each stage of iodoform degradation appears to be important, the extent to which iodoform or diiodomethane are further reduced to iodomethane remains unknown, as it was not possible to quantify in biological matrices with the quantification method used for this study. However, the degradation of iodoform to diiodomethane results in a release of iodine. The transfer of iodine to milk is highly influenced by iodine intake, showing a clear correlation between dietary iodine content in feed and iodine levels in milk (Schöne et al., 2009; Bath et al., 2012). Based on *Asparagopsis* studies, the primary excretion routes for halogens, such as bromine and iodine, are through urine and feces, with some distribution to milk, and limited accumulation in meat, liver, and kidney (Kinley et al., 2020; Stefenoni et al., 2021; Alvarez-Hess et al., 2023, 2024; Cowley et al., 2024). In the present study with iodoform, free iodine concentrations in the biological fluids (digesta, feces, urine, blood, and milk) were not measured. However, Thorsteinsson et al. (2023a) reported in the previously published paper of study I that serum thyroxine (T_4) levels significantly decreased with increasing iodoform doses, indicating increased iodine distribution, although T_4 levels remained within the normal range for a healthy cow.

In a separate experiment (Thorsteinsson et al., 2025), iodoform was mixed into the TMR ranging from 0 to 20 mg/kg DM (0, 8, 16, and 20 mg) of a typical Danish TMR, resulting in a measured total iodine content in milk

ranging from 1.1 to 1.6 mg/kg milk compared with 0.2 mg/kg in milk when the cows did not receive iodoform treatment. The total iodine concentration measured in these milk samples indicates an increased distribution of iodine and iodine-containing compounds to milk when iodoform is added to the diet, although it was not possible to distinguish between the amount of organic or inorganic iodine in these data. The daily iodine intake was ~378 and 515 mg/d for dose 16 and 20 mg, respectively, with a daily milk production of on average 33 kg/d. Based on this information the iodine transfer percentage is ~5.0% and 9.2% of the total iodine intake. Other studies on iodine supplementation have shown 22% to 40% transfer of iodine to milk when fed as a salt (Schöne et al., 2009; van der Reijden et al., 2019). The lower transfer of iodine with iodoform supplementation suggests a low bioavailability of iodine from iodoform compared with dietary iodine in feed rations for dairy cows. It is known that feeding rapeseed and canola meal can significantly lower the transfer of iodine from feed to milk (Franke et al., 2009; Weiss et al., 2015). Rapeseed cake was present in the TMR of the iodoform feeding trials, which also could have affected the iodine transfer from the iodoform treatment. However, a maximum allowed level of 5 mg of iodine per kilogram of total feed as-fed (with 12% water) to dairy cows has been set by the European Commission (Nielsen et al., 2022), which is exceeded 5-fold with the adjusted doses of 20 mg/kg DM of iodoform supplementation in Thorsteinsson et al. (2025). Consequently, regulatory restrictions on iodine content in feed currently hinder the practical use of iodoform as a methane-mitigating additive in dairy production. Iodine is an essential nutritional trace element crucial for synthesizing thyroid hormones such as T_4 and triiodothyronine, which are fundamental for normal growth and metabolism throughout a person's or animal's lifespan (van der Reijden et al., 2017). Nonetheless, excess iodine can prove toxic, leading to hypothyroidism or hyperthyroidism (Roti and Uberti, 2001). The World Health Organization and Food and Agriculture Organization have set a daily recommended intake of 150 µg/d (EFSA Panel on NDA, 2014). Hence, the elevated levels of iodine observed in milk from iodoform-treated cows raise concerns, especially if the milk is intended for various dairy products where the iodine level can be concentrated. The elevated levels of iodine observed in milk from iodoform-treated cows raise concerns, especially if the milk is intended for various dairy products where the iodine level can be concentrated.

Iodoform as a Methane-Mitigating Additive

Reported results by Thorsteinsson et al. (2023a, 2025) demonstrated that dietary provision of iodoform is able to

reduce the enteric methane production from dairy cows. However, considering that iodoform is metabolized to diiodomethane within the initial minutes of exposure in the rumen, it prompts the question of whether diiodomethane is, in fact, the primary methane reducer in cows treated with iodoform. Moreover, it is possible that diiodomethane could potentially replace iodoform entirely as a methane-mitigating feed additive and thereby reduce the release and distribution of iodine into milk compared with the use of iodoform as the additive. However, determining if the replacement of iodoform sufficiently reduces iodine levels requires further investigation.

This study underscores the importance of not only quantifying the parent compound but also examining any resultant metabolites from the administration of halomethane compounds to cows. It is plausible that other organic iodine compounds derived from the metabolism of iodoform or diiodomethane may have been present in milk and urine but were eluded in this investigation. Further investigations are therefore necessary to elucidate the complete ADME profiles of halomethanes considered for methane mitigation purposes, as well as the profile of their degradation products upon ingestion by cattle.

CONCLUSIONS

Iodoform undergoes rapid deiodination to diiodomethane in the rumen with a $T_{1/2}$ of 5.2 min. Diiodomethane can be absorbed across the rumen wall or as rumen content move through the digestive tract, where it is detectable in duodenal fluids. The disappearance of diiodomethane is less rapid in blood serum compared with rumen fluid. Diiodomethane is distributed to milk at high intraruminal doses of iodoform, although with a low systemic availability and distribution in blood and to milk, respectively. Traces of diiodomethane in urine also highlight this route as a possible excretory pathway for diiodomethane.

NOTES

The study was funded by an AU-Arla Fellowship (Aarhus University, Aarhus, Denmark; Agr-2021-731-1418) and a grant from the Innovation Fund Denmark (Aarhus, Denmark; Grant ID: 9067-00024B). Additional funding for the PhD fellowship for M. Rønn was funded by the Graduate School of Technical Sciences (Aarhus University, Denmark). The laboratory technician at AU Viborg (Tjele, Denmark), Birgit H. Løth, is acknowledged for assisting with the iodoform and diiodomethane quantitation analyses. The animal experiments were conducted at Aarhus University Viborg Research Centre Foulum (Tjele, Denmark) and complied with the guidelines set out by the Danish Ministry of Environment and Food with respect to animal experimentation and care of ani-

imals under study (Act No. 2028; Danish Ministry of Environment and Food, 2020). The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: ADME = absorption, distribution, metabolism, and excretion; AUC = area under the curve; $AUC_{0-\infty}$ = AUC from time zero to infinity; AUC_{0-t} = AUC from the last measurable concentration; C_{max} = highest measured concentration; CL = clearance; DIIM = diiodomethane; IF = iodoform; LLE = liquid–liquid extraction; LLOD = lower limit of detection; T_4 = thyroxine; $T_{1/2}$ = half-life; Vd = volume of distribution.

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